Lecture 12: The Unit Cell and the Neutron Life Cycle

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Reading Assignment

Lamarsh & Baratta (3rd or 4th Edition):

- **Section 6.5:** The Four-Factor Formula (The "Engine" of the reactor).
- **Section 6.8:** Heterogeneous Reactors (Why we lump fuel into rods).

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1 Introduction

We have discussed cross-sections (σ) for individual nuclei. Now we must build a machine. Real reactors are not homogeneous mixtures of atoms; they are repeating geometric structures called **Lattices**.

Today we define the "Unit Cell"—the smallest repeating volume of fuel and moderator—and trace the life cycle of a neutron within it. This leads to the fundamental figure of merit for the lattice: the **Infinite Multiplication Factor (k_∞)**.

2 The Neutron Life Cycle (The Four Factor Formula)

Imagine an infinite array of these unit cells. We track a generation of n thermal neutrons absorbed in the fuel.

2.1 1. Fission Production (η)

The cycle begins when thermal neutrons are absorbed by the fuel.

- Not every absorption causes fission (some are captured by ^{238}U).
- η **(Eta):** The average number of fission neutrons produced per *thermal neutron absorbed in the fuel.*

$$n \xrightarrow{\text{Fuel Absorption}} n\eta \quad \text{(Fast Neutrons Born)}$$

2.2 2. Fast Fission (ϵ)

The neutrons are born at high energy (2 MeV). Before they leave the fuel rod, they may strike a ^{238}U nucleus and cause a "fast fission" event.

- This is a small bonus effect.
- ϵ **(Epsilon):** The Fast Fission Factor (> 1). Typically ~ 1.03.

$$n\eta \xrightarrow{\text{Fast Fission}} n\eta\epsilon$$

2.3 3. Slowing Down & Resonance Escape (p)

The neutrons leave the rod and enter the moderator (water). They must slow down from 2 MeV to 0.025 eV.

- **The Danger Zone:** As they pass through intermediate energies (1–1000 eV), they face the giant resonance peaks of ^{238}U. If they touch fuel at this energy, they die.
- p**:** The Resonance Escape Probability. (The chance of surviving the slow-down).

$$n\eta\epsilon \xrightarrow{\text{Slowing Down}} n\eta\epsilon p \quad \text{(Thermal Neutrons Reached)}$$

2.4 4. Thermal Utilization (f)

Now the neutrons are thermal. They diffuse through the lattice. They can be absorbed by the fuel (good) or by the moderator/cladding (bad).

- f**:** The Thermal Utilization Factor.

$$f = \frac{\text{Neutrons Absorbed in Fuel}}{\text{Total Neutrons Absorbed}}$$

$$n\eta\epsilon p \xrightarrow{\text{Diffusion}} n\eta\epsilon p f \quad \text{(New Generation)}$$

2.5 The Result

The ratio of the new generation to the old is:

$$k_\infty = \epsilon p f \eta \tag{1}$$

3 The "Rod" Problem: Why Heterogeneous?

Why do we construct reactors with fuel rods? Why not just dissolve uranium salts in water (Homogeneous)?

The Physics of Lumping:

1. **Resonance Protection (Improving p):** ^{238}U is incredibly "hungry" for neutrons at resonance energies. If you mix U and H atoms uniformly, the neutrons never escape. By lumping the fuel into rods, we force the neutrons to slow down in the *water*, physically separated from the ^{238}U.
2. **Thermal Disadvantage (Hurting f):** Lumping fuel makes it harder for *thermal* neutrons to get back in. The outer skin of the rod "shields" the inside.

4 Design Insight: How Big Should the Rod Be?

A student asked: *"What is the optimal radius for a fuel rod?"* This is a "Goldilocks" problem determined by the Mean Free Path of the neutron.

Quick Recall: The Mean Free Path

Before we optimize the rod size, recall from our discussion on cross-sections that the average distance a neutron travels before colliding with a nucleus is:

$$\lambda = \frac{1}{\Sigma_t} \tag{2}$$

where Σ_t is the macroscopic total cross-section.

- **In Fuel (Fast Neutrons):** Σ_t is small ($\lambda \approx 6$ cm).
- **In Fuel (Thermal Neutrons):** Σ_t is large ($\lambda \approx 1.5$ cm).

The Optimization Constraints

- **Constraint 1: Fast Neutron Escape ($\lambda_{fast} \approx 6$ cm)** Fission neutrons are born fast (2 MeV). They need to escape the rod to reach the water. Since $\lambda_{fast} \approx 6$ cm, a standard 1 cm rod is transparent to them. They fly out easily.
- **Constraint 2: Thermal Neutron Penetration ($\lambda_{thermal} \approx 1 - 2$ cm)** Thermal neutrons must diffuse back *into* the rod to cause fission. Fuel is very absorbent. If the rod is too thick (e.g., 6 cm), the neutrons will all be eaten on the surface, and the center will generate no power.
- **Constraint 3: The Thermal Limit (Melting)** Uranium Oxide (ceramic) is a poor heat conductor. If the rod is too thick, the heat cannot escape, and the centerline will melt ($T_{melt} \approx 2800°$C).

The Solution: Commercial fuel rods are almost universally $\sim$ **1** cm in diameter.

- Small enough for heat and thermal neutrons to get out/in.
- Large enough to provide structural integrity and fuel volume.

5 Cladding: The Critical Shell

[Image of nuclear fuel rod assembly]

5.1 Why do we need it? (The problem with UO_2)

While Uranium Dioxide (UO_2) is the standard fuel due to its high melting point (2865°C) and radiation stability, mechanically and chemically, it is terrible.

- **It is a Ceramic:** Like a dinner plate, it is brittle. Under thermal stress, the fuel pellets crack and fragment.

- **Fission Products:** As fission occurs, the crystal lattice fills with foreign contaminants. Gaseous products (Krypton and Xenon) form bubbles, causing the pellet to swell and potentially crumble.
- **Solubility:** If the hot ceramic comes into direct contact with the coolant (water), it can erode, releasing highly radioactive fission products directly into the turbine loop.

To solve this, we must hermetically seal the fuel pellets in a metal tube known as **Cladding**.

5.2 The Material Search: A "Goldilocks" Problem

Selecting a material for fuel cladding requires balancing two competing requirements:

1. **Nuclear Transparency:** The material must have a low thermal neutron capture cross-section (σ_a) to avoid "poisoning" the chain reaction.
2. **Thermal Resilience:** The material must maintain structural integrity at high operating temperatures (> 300°C) and withstand potential accident conditions.

Table 1: Thermal neutron cross-sections and melting points of potential cladding metals.

Metal	**Absorb.** σ_a **[barns]**	**Melt Point** [°C]	**Verdict / Limitation**
Beryllium (Be)	0.009	1287	Too Brittle & Toxic
Magnesium (Mg)	0.063	650	Corrodes in Water / Low T_m
Zirconium (Zr)	**0.185**	**1855**	**Optimal Balance**
Aluminum (Al)	0.230	660	Melts too easily (Low T_m)
Iron (Fe)	2.560	1538	Parasitic Neutron Absorption
Titanium (Ti)	6.100	1668	Excessive Neutron Absorption

5.3 Analysis of Candidates

The "Winners" that Lost (Be, Mg, Al): Looking at the table, Beryllium and Magnesium are superior to Zirconium in terms of neutron economy. However, Beryllium is mechanically unsuitable (brittle) and hazardous. Magnesium (used in early gas-cooled reactors) and Aluminum cannot withstand the high temperatures and corrosive environment of modern water-cooled reactors; their melting points are dangerously close to accident temperatures.

The Iron Problem: Iron (steel) is structurally excellent but acts as a "neutron hog." With a cross-section of 2.56 barns (over $10\times$ that of Zirconium), using steel cladding would require significantly higher uranium enrichment to compensate for the lost neutrons, making the fuel cycle uneconomical.

Why Zirconium Wins: Zirconium is the only candidate that provides "good enough" neutron transparency (≈ 0.185 b) while possessing a melting point high enough (1855°C) to serve as a safety barrier.

5.4 The Hafnium Contamination Issue

There is one major catch. In nature, Zirconium is always found combined with **Hafnium** (Hf). They are "chemical twins" (located in the same column of the periodic table) and are extremely difficult to separate.

- **Zirconium:** $\sigma_a = 0.185$ barns (Transparent).
- **Hafnium:** $\sigma_a \approx 104$ barns (Black).

Natural Zirconium typically contains 1–3% Hafnium. If you built a reactor out of "Commercial Grade" Zirconium, the Hafnium impurity would poison the reaction, making criticality impossible.

This leads to the distinction between two grades of metal:

- **Commercial Grade Zr:** Contains Hf. Used in chemical plants. Cheap.
- **Nuclear Grade Zr:** Hf removed via difficult liquid-liquid extraction. Expensive, but required for cladding.

Ironically, the separated Hafnium is so absorbent that we use it to make the Control Rods!

Historical Note: The Zirconium-Hafnium Miracle

The use of Zirconium in nuclear reactors was not obvious; it was forced into existence by Admiral Hyman G. Rickover for the *USS Nautilus*.
In the late 1940s, published data suggested Zirconium had a high neutron cross-section, making it unsuitable for reactors. Researchers at MIT and Oak Ridge National Laboratory (ORNL) suspected this data was flawed and discovered the culprit: the inevitable 2% Hafnium impurity found in all natural Zirconium ore.
When Rickover was told that separating the two "chemical twins" was commercially impossible, he refused to accept the answer. He drove the development of a liquid-liquid extraction process (using methyl isobutyl ketone) at the Y-12 plant in Oak Ridge. He then forced the Bureau of Mines and the Wah Chang Corporation to scale the process from grams to tons. This effort dropped the price of reactor-grade Zirconium from over $300/lb to under $5/lb, making the nuclear navy possible.

Reference: Rickover, H. G. (1975).*History of the development of zirconium alloys for use in nuclear reactors*. U.S. Energy Research and Development Administration, Division of Naval Reactors.

6 Historical Sidebar: The Natural Reactors of Oklo

You might think that a nuclear reactor requires precise engineering to separate fuel and moderator. In 1972, French physicists discovered that nature beat us to it by 1.7 billion years.

The Oklo Phenomenon (Gabon, Africa): Mining engineers found uranium ore that was depleted in ^{235}U (found at $\sim 0.4 - 0.6\%$ instead of the standard 0.72%). They traced it back to a deposit that had acted as a natural 100 kW nuclear reactor for hundreds of thousands of years.[1]

How did it work?

[1] For more information, read the Wikipedia article: Natural nuclear fission reactor

- **Higher Enrichment:** 1.7 billion years ago, the natural abundance of ^{235}U was $\sim 3\%$ (because it decays faster than ^{238}U). This is the same enrichment we use in modern PWRs!
- **Separation of Phases:** The uranium ore was concentrated in veins, and groundwater flooded the cracks and pores in the rock.
- **The "Lattice":** The rock provided the fuel; the water in the cracks provided the moderator. The heterogeneous separation allowed p to be high enough for criticality.
- **Passive Safety:** When the reactor got too hot, it boiled the water away. Without the moderator, the reaction stopped (negative void coefficient). When the rock cooled, water returned, and the reactor restarted.